

LUCKY SAH

Robotics Researcher | Mechanical Engineering | Electromechanical Prototyping

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PROFESSIONAL SUMMARY

Undergraduate robotics researcher and full-stack robotics developer pursuing a degree in mechatronics engineering. Develops humanoid robotic systems from mechanical design and fabrication through hardware integration, robotics software, and simulation, using SolidWorks, Autodesk Inventor, ROS, Isaac Sim, and MuJoCo. Coauthored research papers address mechanical systems, agentic robotics, and human-robot interaction.

EXPERIENCE

Robotics Researcher

[ADD START DATE] – Present

Advanced Telerobotics Research (ATR) Lab, Kent State University

- Lead mechanical engineering workstreams and prototype development for robotics research, advancing systems from parametric CAD through fabrication, assembly, integration, and functional hardware.
- Own mechanical workstreams spanning structures, mounts, mechanisms, actuator integration, sensor packaging, and embedded hardware integration.
- Manage prototyping resources, equipment, workspace, and hands-on engineering support across interdisciplinary research teams.

Robotics Hardware Intern (Co-op)

[ADD DATES]

PlaiPin Inc.

- Translate parametric CAD concepts into build-ready mechanical assemblies and functional prototypes for a modular AI companion robot.
- Design and iterate structural components, mounts, and actuator interfaces while supporting fabrication, assembly, hardware integration, and prototype refinement.

Automation and Product Development Intern

2022–2024

Thoplo Machine Pvt. Ltd. & Baliyo Ventures Pvt. Ltd.

- Contributed to the mechanical design, fabrication, assembly, and testing of automated production machinery and supporting systems.
- Supported PLC lab setup and wiring, industrial equipment assembly, machining, fiber-reinforced molding, and production-focused fabrication.

SELECTED PUBLICATIONS

I. S. Cardenas, M. A. Arnett, N. C. Yeo, **L. Sah**, and J.-H. Kim, "ROSClaw: An OpenClaw ROS 2 Framework for Agentic Robot Control and Interaction," *arXiv preprint*, arXiv:2603.26997, 2026.

M. Chen, **L. Sah**, A. Rivera, and D. Morgan, "Design and Experimental Evaluation of a Low-Cost Modular Mobile Robot for Human-Robot Interaction Research," *Proceedings of the Sample Robotics Research Conference*, pp. 112–119, 2026. (*fictional layout sample*).

A. Rivera, M. Chen, **L. Sah**, and D. Morgan, "A Compact Tendon-Driven Actuation Mechanism for Expressive Humanoid Robotic Heads," *Proceedings of the Sample Symposium on Humanoid Robotics*, pp. 48–55, 2026. (*fictional layout sample*).

SELECTED ENGINEERING PROJECTS

Mini TurtleBot — Low-Cost Modular Mobile Robot | ATR Lab

- Designed a chassis and motion architecture around low-cost fabrication, modular expansion, and straightforward integration of robotic hardware. [ADD TEST RESULT OR COST]

KAMI Head — Human-Robot Interaction Module

- Designed an expressive robotic head assembly combining mechanical motion systems and embedded components for human-robot interaction. [ADD DEGREES OF FREEDOM OR DEMO OUTCOME]

TECHNICAL SKILLS

Mechanical Design: Autodesk Inventor Professional, Fusion 360, SolidWorks, parametric assemblies, mechanisms, 3D modeling, bill-of-materials development

Robotics & Embedded: ROS, Arduino, microcontrollers, actuators, sensors, motor drivers, electromechanical integration

Fabrication: FDM and resin 3D printing, CNC machining, soldering, power tools, silicone, polyurethane, and fiber-reinforced molding

Automation: PLC fundamentals, industrial equipment assembly, wiring, and production-oriented fabrication

EDUCATION

B.S. in Mechatronics Engineering; Minor in Computer Science

Expected 2028

Kent State University — Kent, Ohio

INTERNATIONAL ACTIVITIES

Selected competitions & challenges: World Robot Olympiad (Thailand); FIRST Global Challenge (Switzerland); Asian Development Bank Hackathon; Robo-Battle Bots; Robo Soccer. [ADD ROLE, YEAR, AND RESULT]

Selected conferences & programs: Founders, Inc. (San Francisco); Asian Pacific Conference of Young Scientists. [ADD ROLE]